

Question:

A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

Answer Choices:

- A. Epstein-Barr virus (EBV)
- B. Measles virus
- C. Human immunodeficiency virus (HIV)
- D. Human papillomavirus

Reasoning + Answer:

<think>

- CD 2 1 is a receptor for EB V and plays a crucial role in the B cell response to this virus .

- EB V uses the CD 2 1 receptor to enter and infect B cells , which is an essential part of its pathogenesis .

- Other viruses like measles , HIV , and human pap ill om avirus do not primarily use CD 2 1 for infection .

- Therefore , modifying B cells to not express CD 2 1 would most significantly impact the infection and pathogenesis of EB V .

- This means that the path ogenesis of EB V would be most affected by this genetic modification .

- The other viruses listed do not use CD 2 1 as a receptor .

- Hence , the correct answer is the one related to EB V .

- EB V is the only virus that requires CD 2 1 for infection .

- The genetic modification of B cells to not express CD 2 1 would directly interfere with EB V's ability to infect and replicate .

- This interference would most significantly affect the path ogenesis of EB V .

- The answer choices are given , and the correct answer is the one related to EB V .

- The correct answer is the one that corresponds to EB V .

- The correct answer is A , as EB V uses CD 2 1 for infection .

- The other viruses listed do not use CD 2 1 as a receptor .

- Therefore , the genetic modification of B cells to not express CD 2 1 would most significantly affect the path ogenesis of EB V .

- The correct answer is A , as EB V uses CD 2 1 for infection .

- The other viruses listed do not use CD 2 1 as a receptor .

- Therefore , the genetic modification of B cells to not express CD 2 1 would most significantly affect the path ogenesis of EB V .

</think>

< answer >A . Epstein -B arr virus (EB V)</ answer >